

DIANE MALINDA SALIM

diane.salim@rutgers.edu

PERSONAL INFORMATION

Research Interests

- Star formation
- Galaxy Evolution
- Interpretable Machine Learning
- Interstellar Medium
- Turbulence
- Numerical Simulations

Proficient computer languages

- Python, Interactive Data Language (IDL), $\text{\LaTeX}$

Additional advanced business-proficient languages

- Japanese (JLPT N2, DipLang), Indonesian

EDUCATION

Rutgers University

Doctor of Philosophy in Physics and Astronomy

Master of Philosophy in Physics and Astronomy

September 2020 - present
New Brunswick, United States of America

Australian National University

Bachelor of Philosophy (Science) with First Class Honours

Diploma of Languages in Japanese

Januray 2014 - December 2018
Canberra, Australia

The University of Tokyo

University-wide Student Exchange Program (USTEP)

April-July 2016; April - July 2017
Tokyo, Japan

Monash University

Extension studies in chemistry

January 2013 - December 2013
Melbourne, Australia

John Monash Science School

Australian Tertiary Admissions Rank (ATAR) of 99.45

January 2011 - December 2013
Melbourne, Australia

WORK EXPERIENCE

Flatiron Institute Center for Computational Astrophysics

Guest Researcher

July 2020 - present
New York, United States of America

GRID Inc.

Machine Learning Engineer for Computer Vision

June 2019 - May 2020
Tokyo, Japan

Max Planck Institute for Astronomy

Summer Scholar

September 2017 - December 2017
Heidelberg, Germany

Japan Aerospace Exploration Agency

New Colombo Plan Scholar and Technical Trainee

October 2016 - February 2017
Sagamihara, Japan

National Astronomical Observatory of Japan

New Colombo Plan Scholar and Intern

February 2016 - March 2016
Tokyo, Japan

Carnegie Observatories

Summer Scholar

November 2015 - December 2015
Pasadena, United States of America

REFEREED PUBLICATIONS

As first
author:

1. *A data-driven approach for star formation parameterization using symbolic regression*
Salim, D. M.; Orr, M. E.; Burkhardt, B.; Somerville, R.; Cranmer, M.; 2025, The Astrophysical Journal, 995, 73.
2. *Extending a Physics-Informed Machine Learning Network for Superresolution Studies of Rayleigh-Bénard Convection*
Salim, D. M.; Burkhardt, B.; Sondak, D. 2024, The Astrophysical Journal, 964, 2.
3. *Spinning Bar and a Star-formation Inefficient Repertoire: Turbulence in Hickson Compact Group NGC7674*
Salim, D. M.; Alatalo, K.; Federrath, C.; Groves, B.; Kewley, L. K. 2020, The Astrophysical Journal, 893, 26.
4. *A Universal, Turbulence-Regulated Star Formation Law*
Salim, D. M.; Federrath, C.; Kewley, L. J. 2015, The Astrophysical Journal Letters, 806, L36

As second author: (Significant contributions)	1. <i>On the effective turbulence driving mode of molecular clouds formed in disc galaxies</i> Jin, K.; Salim, D. M. ; Federrath, C; Tasker, E. J.; Habe, A.; Kainulainen, J. T. 2017, Monthly Notices of the Royal Astronomical Society, 469 (1): 383-393. 2. <i>The SAMI Galaxy Survey: a new method to estimate molecular gas surface densities from star formation rates</i> Federrath, C.; Salim, D. M. ; Medling, A. M. et al. 2017, Monthly Notices of the Royal Astronomical Society, 468 (4): 3965-3978.
Other publications:	1. <i>Characterizing the Molecular Gas in Infrared Bright Galaxies with CARMA</i> Alatalo, K.; ... Salim, D. M. et al. 2024, The Astrophysical Journal Letters, 975, 241

SELECTED AWARDS AND HONOURS

Quad Fellowship	<i>June 2023 - May 2024</i>
- Initiative of the governments of the Quad countries (Australia, India, Japan, USA) to sponsor graduate students for study in the USA with the mission to elevate the brightest minds in science for social good. - One of 100 students selected to receive inaugural personal award of \$50,000.	
Future Investigators in NASA Earth and Space Science and Technology (FINESST) in Astrophysics Fellowship	<i>June 2022 - May 2025</i>
- Award to funds graduate student research projects that best contribute to NASA Science Mission Directorate's science, technology, and exploration goals. - One of 29 proposals accepted with an acceptance rate of 13% and total personal funding award of \$150,000.	
Rutgers University Physics Department Fellowship	<i>January 2021 - December 2021</i>
- Award for the top-ranking applicant to Rutgers University Department of Physics Doctor of Philosophy Program. - Funds one independent year of study in the doctoral program.	
New Colombo Plan Scholarship to Japan	<i>January 2016 - September 2017</i>
- Australian Government Department of Foreign Affairs and Trade initiative to sponsor Australian undergraduate students to study abroad and intern in Asian for up to 19 months to develop their Indo-Pacific capability and Asia literacy. - One of 100 students selected annually to receive award of \$2,500 per month of in-country study and employment.	

SELECTED INTERNATIONAL CONFERENCES AND SEMINARS

ORAL PRESENTATION: A data-driven approach for star formation parameterisation	
• The Magnetized Turbulent Universe (<i>Invited, LOC</i>)	<i>November 2025 - Playa Del Carmen, Mexico</i>
• UniversAI: Exploring the Universe with Artificial Intelligence	<i>June 2025 - Athens, Greece</i>
• Swinburne University Center for Astrophysics and Super-computing Colloquium (<i>Invited</i>)	<i>August 2024 - Melbourne, Australia</i>
• Machine Learning for Astrophysics 2 nd edition	<i>July 2024 - Catania, Italy</i>
• Star Formation across cosmic scales: Machine Learning insights and applications	<i>May 2024 - Budapest, Hungary</i>
ORAL PRESENTATION: A physics-informed superresolution model for turbulence	
• Galaxy Formation and Evolution in the Data Science Era	<i>March 2023 - Santa Barbara, United States</i>
• From Stars to Galaxies II	<i>June 2022 - Gothenburg, Sweden</i>
• European Community on Computational Methods in Applied Sciences 2022	<i>June 2022 - Oslo, Norway</i>
ORAL PRESENTATION: A turbulence-regulated star formation law	
• The laws of star formation: From Cosmic Dawn to Present Universe	<i>July 2018 - Cambridge, United Kingdom</i>

REFEREES

Blakesley Burkhart (Rutgers University and Flatiron Institute)	bburkhart@flatironinstitute.org
Rachel Somerville (Flatiron Institute)	rsomerville@flatironinstitute.org
Romeel Davé (The University of Edinburgh)	romeel.dave@ed.ac.uk